

Estimating Value-at-Risk of a Portfolio Using GARCH-Family Models

This project estimates Value-at-Risk (VaR) for an equally-weighted portfolio using GARCH-family volatility models.

Portfolio Returns Computation

For the purpose of analysis, a portfolio consists of 5 assets is built. Five daily price series are pulled from Yahoo Finance for the period May 2020 – May 2025 using *yfinance*: AAPL, S&P 500 (^GSPC), USD/CNY, WTI Crude (CL=F), and ETH-USD. For each asset, log-returns are computed, the equally-weighted portfolio return is then the row-wise mean of the five log-returns.

Stylised Facts in Volatility Modelling

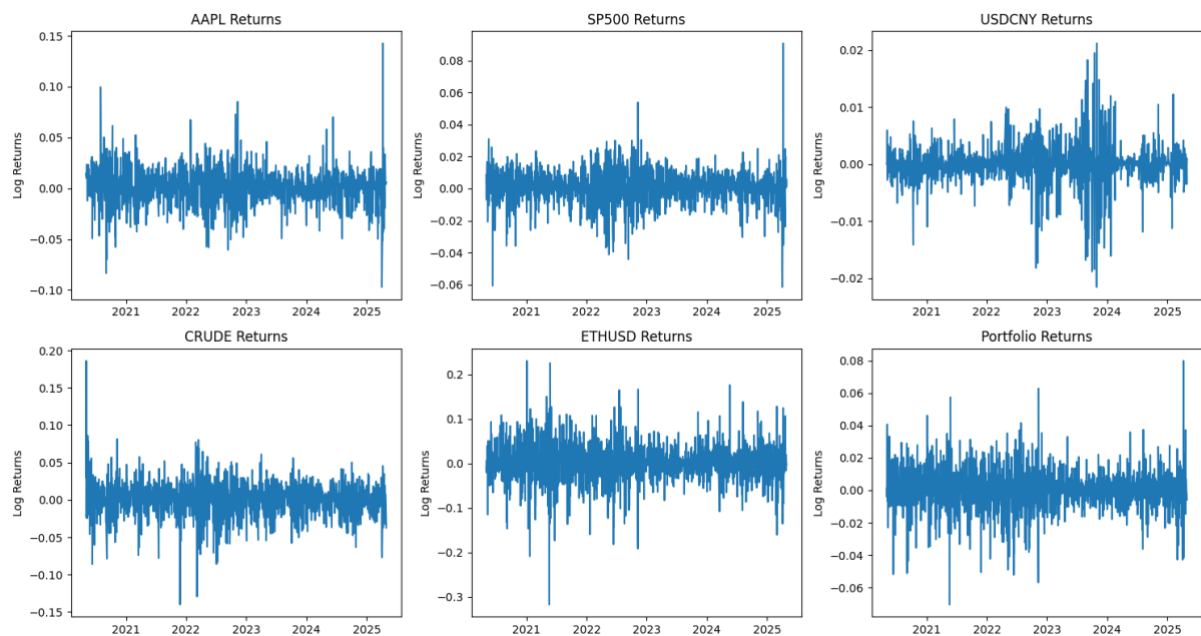


Figure 1. Portfolio and assets daily returns

Figure 1 presents daily returns of each asset and the portfolio. ‘Volatility clustering’ can be observed as the amplitude of return varies over time — periods of large returns (in absolute value) tend to be followed by further large returns, and calm periods cluster similarly.

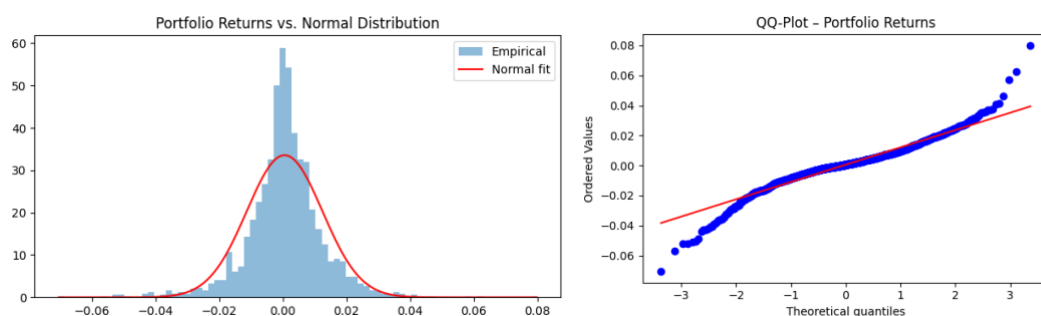


Figure 2. Portfolio returns compared with normal distribution

The JB statistic of 1557.45 (p-value ≈ 0) rejects the null hypothesis of normality. The empirical distribution shows a taller, narrower central peak and visibly heavier tails compared to the fitted normal curve. QQ-plot shows evidence of heavy tails in distributions with points in the upper-right tail rise above the line and points in the lower-left tail fall below the line.

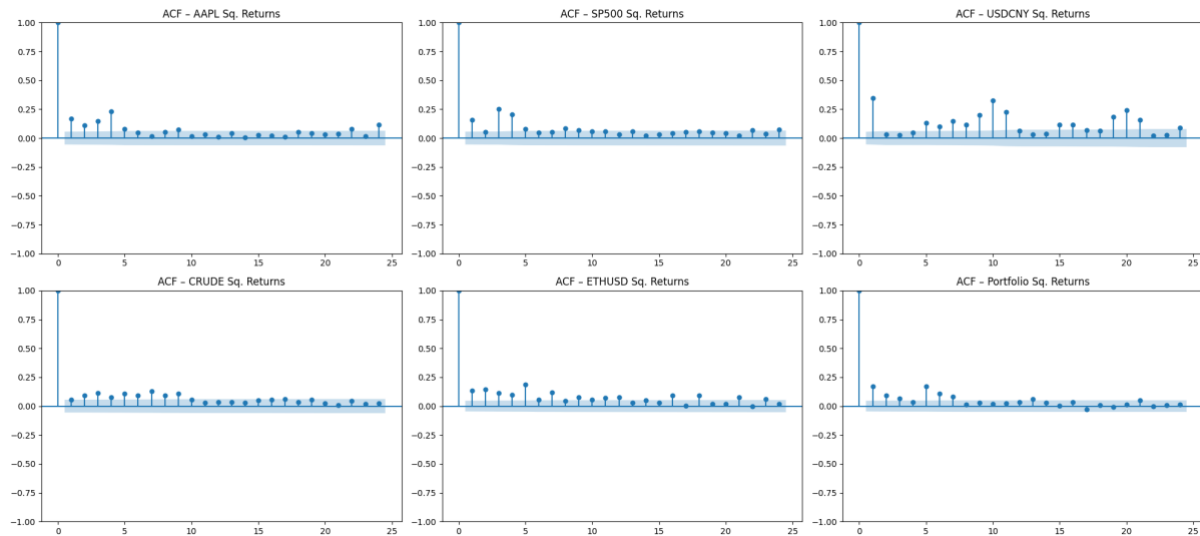


Figure 3. ACF of squared returns

The portfolio shows evidence of ARCH effects as presented by the autocorrelations of squared returns in Figure 3. A LM test statistics of 111.761 (p-value ≈ 0) also rejects the “no ARCH” hypothesis.

Empirical observation of these characteristics supports the use of GARCH-family models for volatility and risk estimation.

GARCH Model Estimation

Six specifications are estimated in order of increasing complexity: ARCH(1) $\rightarrow$ ARCH(2) $\rightarrow$ ARCH(5) $\rightarrow$ GARCH(1,1) $\rightarrow$ GARCH(2,1) $\rightarrow$ GARCH(1,2), plus EGARCH(1,1). All models are fit to in-sample portfolio log-returns (May 2020 – Apr 2024). For each model, diagnostic tools are applied to the standardized residuals.

Model	Ljung-Box Test on Standardized Residuals lag 5	Ljung-Box Test on Standardized Residuals lag 20	Ljung-Box Test on Standardized Squared Residuals lag 5	Ljung-Box Test on Standardized Squared Residuals lag 20	LM-ARCH (5 lags)
ARCH(1)	0.295	0.292	0.000	0.000	0.000
ARCH(2)	0.429	0.341	0.000	0.000	0.000
ARCH(5)	0.616	0.638	0.951	0.087	0.952
GARCH(1,1)	0.627	0.718	0.264	0.372	0.217
GARCH(2,1)	0.637	0.717	0.086	0.219	0.064
GARCH(1,2)	0.605	0.687	0.030	0.124	0.020

Table 1. Diagnostics Summary — p-values for standardized residuals

Model	AIC	BIC
ARCH(1)	-8789	-8774
ARCH(2)	-8809	-8788
ARCH(5)	-8856	-8819
GARCH(1,1)	-8922	-8901
GARCH(2,1)	-8917	-8890
GARCH(1,2)	-8924	-8897

Table 2. AIC / BIC comparison

GARCH(1,1) is the preferred specification as it passes all the residual diagnostics, has the best BIC among all models.

In Ljung-Box test on standardized residuals, all p-values are above 0.05. We fail to reject H_0 of no serial correlation in standardized residuals at all lags. The mean equation (constant mean) is correctly specified — no autocorrelation structure in returns is left unexplained.

In Ljung-Box test on standardized squared residuals, all p-values comfortably above 0.05. We fail to reject H_0 of no autocorrelation in squared residuals. GARCH(1,1) has successfully absorbed the volatility dynamics — squared residuals is white noise, meaning no predictable variance structure remains in the residuals.

In LM-ARCH test, $p = 0.217$ means we fail to reject H_0 of no remaining ARCH effects.

GARCH(1,1) is a well-specified volatility model for the portfolio. The standardized residuals behave as white noise in both levels and squares.

Parameters Interpretation

Parameter	Estimate	p-value	Expected sign	Typical range
mu	7.136e-04	9.878e-07	> 0	small positive
omega	2.892e-06	0.00	> 0	small positive
alpha1	0.0500	3.080e-06	> 0	0.05 – 0.15
beta1	0.9300	0.000	> 0	0.85 – 0.95

Table 3. Estimated GARCH(1,1) parameters (in-sample, 1460 observations)

In the estimated GARCH(1,1) model, all four parameters are statistically significant, signs are correct.

Mu is the average daily portfolio log-return.

Omega is the long-run contribution to variance. The annualised unconditional long-run variance implied by the model approx. 19%.

Alpha measures how much a large return shock at $t-1$ immediately elevates today's variance. A value of 0.05 indicates shocks have a moderate immediate impact.

Beta Measures how much of yesterday's conditional variance carries over to today. At 0.93, volatility is highly persistent — a shock to variance takes a long time to decay back to the unconditional level.

A sum of alpha and beta is less than 1, matches the requirement as a mean reverting variance process.

Parameter	Estimate	p-value
mu	5.96e-04	0.042
omega	-0.0150	0.462
alpha1	0.0588	0.003
gamma1	-8.0e-05	0.994
beta1	0.9979	0.000

Table 4. Estimated EGARCH(1,1) parameters (in-sample, 1460 observations)

For the specified EGARCH(1,1) model, estimated gamma is insignificant, meaning no statistically detectable asymmetry, positive and negative shocks of equal size have the same impact on volatility.

Parameter	Estimate	p-value
mu	5.695e-04	0.017
omega	-0.0110	0.492
alpha1	0.0516	0.000
gamma1	-0.1188	0.010
gamma2	0.1218	0.009
beta1	0.9985	0.000

Table 5. Estimated EGARCH($p=1, o=2, q=1$) parameters (in-sample, 1460 observations)

However the specified EGARCH($p=1, o=2, q=1$) model suggests both immediate leverage effect at lag 1 and a delayed effect at lag 2.

Diagnostics Interpretation

Nyblom stability test

Joint Statistic: 239.414

Individual Statistics:

omega 32.8162

alpha1 0.1107

beta1 0.1380

Asymptotic Critical Values (10% 5% 1%)

Joint Statistic: 0.846 1.01 1.35

Individual Statistic: 0.35 0.47 0.75

Nyblom parameter stability test – omega is unstable, the baseline variance level shifts across regimes, the variance floor is not constant over in-sample period. Joint statistic is also rejected driven by omega.

Sign Bias Test

t-value prob sig

Sign Bias 0.03379 0.973049

Negative Sign Bias 2.97066 0.003010 ***

Positive Sign Bias 0.53283 0.594215

Joint Effect 11.78392 0.008161 ***

Sign bias test suggest that negative shocks have a larger effect than GARCH(1,1) captures, standard GARCH(1,1) misses the leverage effect — negative portfolio shocks elevate volatility more than positive shocks of equal magnitude. Joint effect statistics also confirms overall asymmetry.

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	174.8	3.236e-27
2	30	206.3	1.186e-28
3	40	201.1	1.045e-23
4	50	221.7	8.296e-24

Adjusted Pearson Goodness-of-Fit test results in all four groupings reject normality. Even after GARCH(1,1) removes the volatility clustering, the standardized residuals are still not normally distributed. Under a normal assumption, model can underestimate tail risk, as the distribution observes a heavier tail than a standard normal.

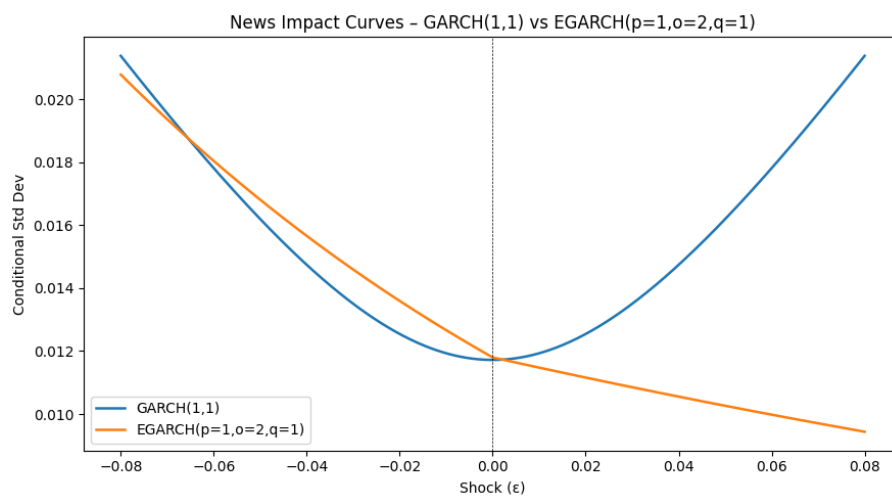


Figure 4. News Impact Curves of GARCH and EGARCH model estimated

GARCH curve shows a standard symmetric U-shape, identical response to positive and negative shocks. While EGARCH curve is asymmetric — negative shocks produce higher conditional volatility than positive shocks of equal magnitude, which is the leverage effect captured by this model.

Value-at-Risk Calculation

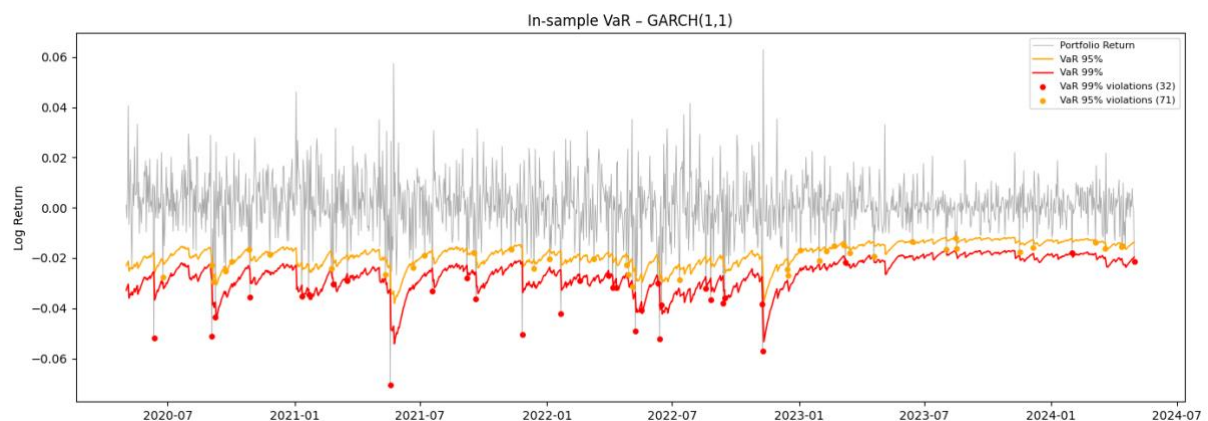


Figure 5. In-sample value-at-risk by GARCH(1,1)

In-sample VaR is evaluated on the same data used for estimation. At a 95% level, Expected violations: 73. Observed: 71. The observed count is well within one standard deviation of the expected — the model is correctly calibrated at the 95% level. At a 99% level, Expected violations: 15. Observed: 32. The model is systematically underestimating the probability of very large losses.

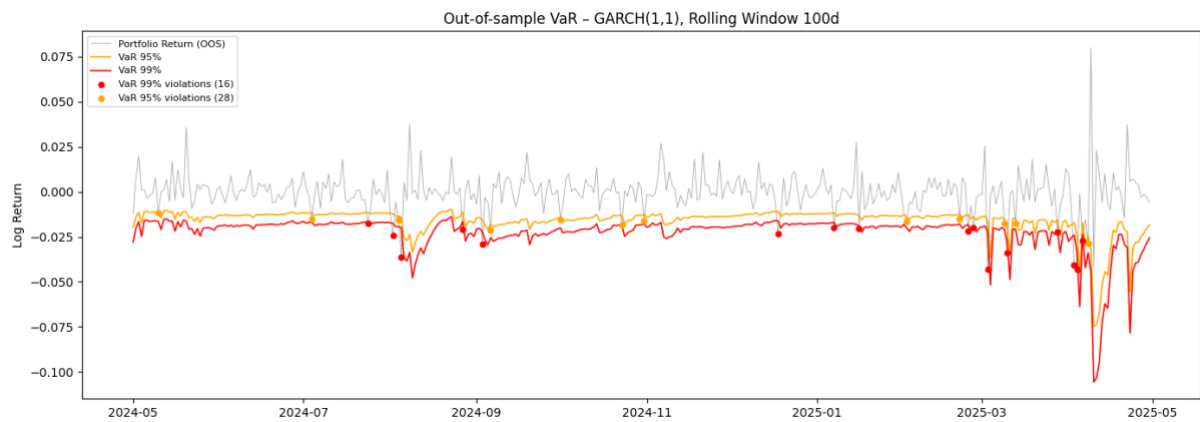


Figure 6. Out-of-sample value-at-risk by $GARCH(1,1)$

Out-of-sample value-at-risk is calculated using a rolling forward method with a 100-day window. At a 95% level, Expected violations: 18.3. Observed: 28. At a 99% level, Expected violations: 3.65. Observed: 16. The model underestimates both tail distribution and level of volatility in out-of-sample period. Possible cause is the volatility spike at the end of out-of-sample period and high persistence in the model responds slowly to the spike.